EXPECTED VALUE

Which properties need what

Linearity needs nothing; the product rule needs independence; the mean itself must exist.

✓ ALWAYS · ◐ IF INDEPENDENT · ● IF IT EXISTS

Linearity — ✓ ALWAYS

$$E(X+Y) = E(X) + E(Y)$$

even when X and Y are dependent

Scaling and shifting — ✓ ALWAYS

$$E(aX+b) = a\,E(X) + b$$

changes of units behave predictably

A constant — ✓ ALWAYS

$$E[c] = c$$

Iterated expectation — ✓ ALWAYS

$$E\big[E[X \mid Y]\big] = E[X]$$

averaging the conditional mean gives the mean

Product — ◐ IF INDEPENDENT

$$E(XY) = E(X)\,E(Y)$$

X, Y independent

does not hold under dependence

Existence — ● IF IT EXISTS

the sum or integral must converge

the Cauchy distribution has no finite mean

Toss a coin until the first head; a head on toss k pays 2^k dollars. Every